False memories to fake news: The evolution of the term “misinformation” in academic literature

Rough outline

State of the misinformation paradigm

- History as currently told
- Distinguish phenomenon and paradigm
- What's missing?
- Definition of the paradigm

History of the paradigm

- Data and methods
- Find what's missing
- Historical context

Theoretical critique

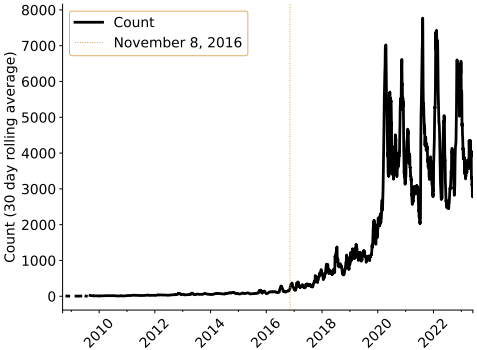
- Comparison to alternative framings
- Theoretical and practical shortcomings

Conclusion

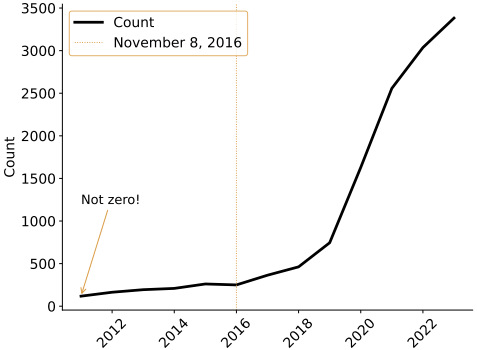
- Some soapboxing[1]
- Ideas for future work

[1] As opposed to the rest of this talk, which will be *nonpartisan* and *apolitical*

“Misinformation”



"Misinformation" on Twitter



"Misinformation in academic literature"

Brief anatomy of the paradigm

Definition

- There are false statements, i.e., *misinformation*
- People make, read, and share misinformation, often (usually?) on social media
- The sum of many such events results in harm to society

Example supplementary concepts

- Disinformation
- Fact-check
- Infodemic
- Fake news
- Faith in institutions
- (Anti-)science

Notably missing

- Epistemology
- Mediation

How misinformation studies tells its history

General summary

- Misinformation has always existed because people have always said false things
- Technology affects the spread of misinformation
- Social media lead to some kind of inflection point

Examples from two lit reviews

- "While history shows that false and misleading information is not a new phenomenon, most observers seem to agree that misinformation, disinformation, and fake news have become much more prevalent during the last decade." [1]
- "Although misinformation can circulate exceedingly fast due to advances in social technologies and large-scale information cascades, the roots of fake news go back to the days before the Printing Revolution, when word-of-mouth was the primary method of news transmission." [2]

What's missing?

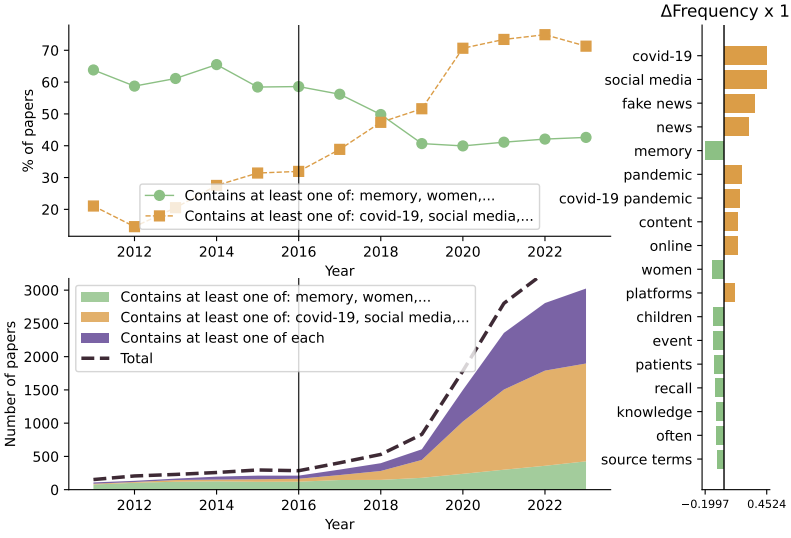
- *Misinformation* is a scientific paradigm [3] with more conceptual baggage than its literal definition
- Histories are told within the paradigm, not of the paradigm
- Where does the post-2016 concept come from? Two options: creatio ex nihilo vs creatio ex materia
- An explanation of pre-2016 term use



Term frequency

- Scopus [4] export of all papers with “misinformation” in any metadata field from 2011–2023
- Term frequencies before/after 2016

Term frequency



History of the term in the literature

50s–60s: Medicine and Public Health

- 1956: “Combating Food Misinformation and Quackery” [5]
- 1957: “Information and Misinformation Gained from Fasting Blood Sugar Alone in Diabetes Therapy”[6]

60s–70s: Public health dominates, but usage diversifies

- 1970: “A Model of the Soviet Firm” discusses Soviet firms’ attempts to correct for “systematic misinformation” by changing incentives [7]

70s–90s: More specialized uses with technical meanings, e.g. associated with computation

- **1974: “Reconstruction of automobile destruction: An example of the interaction between language and memory” introduces the “misinformation effect”[8]**
- 1989: “Categorical Approach to Distributed Systems, Expressibility and Knowledge:”It is asserted that to obtain a meaningful comparison of protocols, the adversary should be chosen not to send identical faulty messages but to ‘convey’ the same misinformation” [9]

History of the term in the literature

2000s

- 2002: “Infodemiology: The Epidemiology of (Mis)Information” studies “the study of the determinants and distribution of health information and misinformation”[10]
- 2007–2010 - Infodemiology extends to study how (mis)information spreads during crises (earthquakes, epidemics, etc.) on social media [13]

2010s to today

- The *post-2016 paradigm* is born

Creatio ex nihilo vs creatio ex materia

Methodology

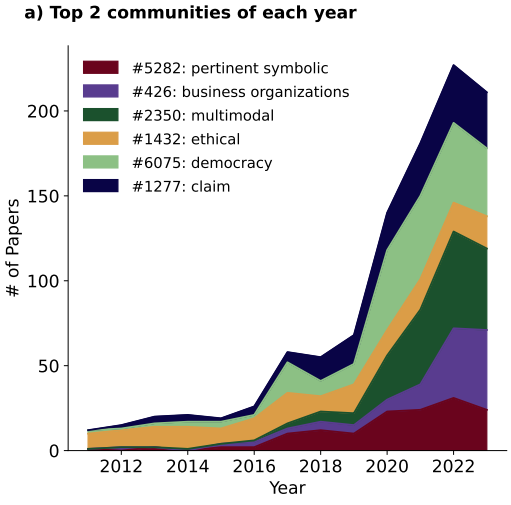
- Louvain community detection [14]
- Papers as nodes; authors as edges
- Reduce weight of the edges of big nodes

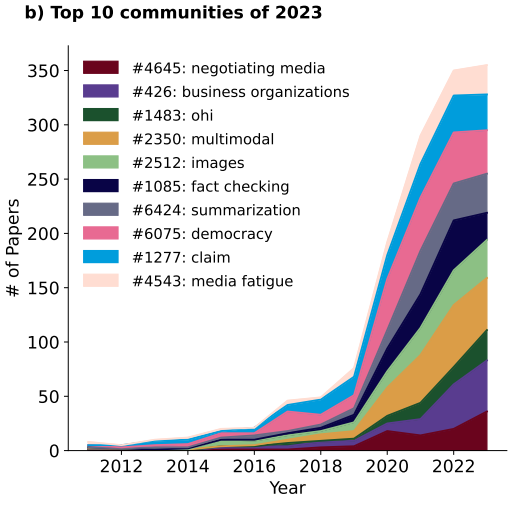
$$W(P_1, P_2) = \frac{1}{len(P_1.\text{authors}) \times len(P_2.\text{authors})}$$

Two interpretations

- The top ten communities of 2023 through all time
- The top two communities of every year through all time

Creatio ex nihilo vs creatio ex materia

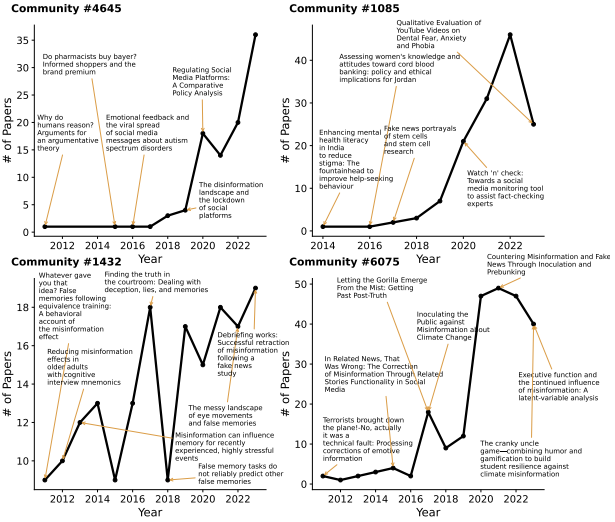




TF-IDF

Community	TF-IDF (unigram-trigram)
5282	pertinent symbolic, misleading content, symbolic, pertinent, content based, content, language, linguistic, representative, detection, real world, accuracy also, additional training, aggregating representative, also time, alternative multi, analyzes linguistic, attributes characterize, based analytical, board showing
426	business organizations, media governments, news media, pandemic research, governments, pandemic, health crises, trust, crises, business, arise 2024, building unbuilding, businesses communication, chapter aimed, chapter proposed, competent curbing, compliance stopping, crises arise, crises public, curbing global
2350	multimodal, dis, multimodal analysis, multimodal dis, disciplines, review, computer science, computer, science, analysis, gaps, studies, research, communication, 2020 start, analysis dis, attention academics, beyond words, bird eye, brings research
6075	democracy, attacks involves, campaigns undermining, citizens example, contribute countermeasures, countermeasures 2025, democracy democracy, democracy relies, discredited recent, elections reliable, epistemic integrity, evidence widespread, example trust, frontier attacks, involves researchers, knowledge establish, list several, pattern science, psychology contribute, recent frontier
1277	claim, claim task, task, claims, social media, spread false, false, media, participation, social, fake news, 2023 forum, 2023 primary, 2023 uncovering, 28 teams, 40 registrations, age 2024, become spread, behemoths, behemoths hire
1432	ethical, debriefing, ethical practices, ethical research, researchers, research, dialogue, deception, adopting reporting, balance deception, benefits individual, challenges insurmountable, conducting ethical, consent value, consider three, debriefing make, debriefing research, deception dialogue, deception informed, dialogue debriefing

Solidification of the paradigm

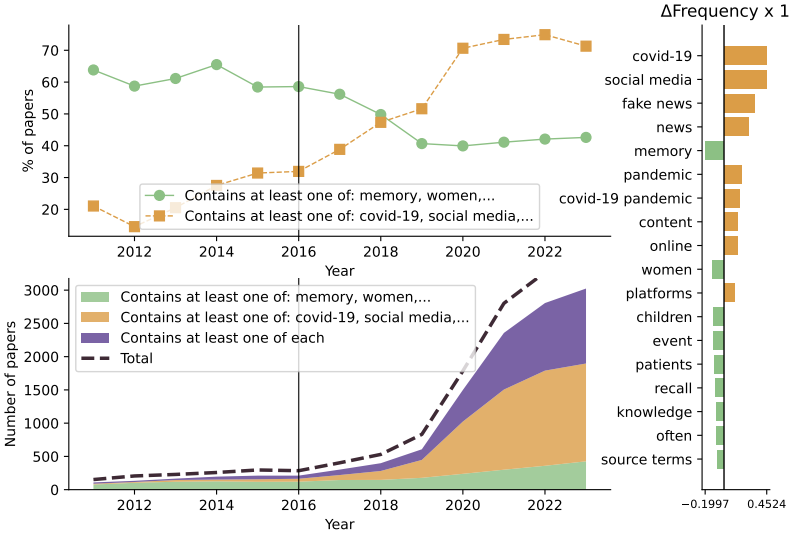


Random example from #1432 (aka Loftus community)

Protecting Against Misinformation: Examining the Effect of Empirically Based Investigative Interviewing on Misinformation Reporting [15]

- "Children who are involved in legal cases are often interviewed about events they witnessed or that might have happened to them... We found that children's recall during the NICHD interview protected children against the incorporation of misinformation in their accounts of the event ..."
- Published in *Journal for Police and Criminal Psychology*

This explains our term frequency changes!



Context: Satanic panic

1980: *Michelle Remembers*

- Michelle Smith co-authors Michelle Remembers with her psychiatrist and soon-to-be-husband, Lawrence Pazder, about her recovered memories of a childhood of abuse at the hands of a Satanic cult.
- Kicks off media circus and marks the beginning of the mainstream Satanic panic [16] [17].

1983–1990: McMartin preschool trials [18] [19] [20] [21]

- A single parent accuses ex-husband and school staff member of child abuse.
- Police send letter to 200 parents at the school saying that their children might have been victims of sexual abuse, outlining the potential abuse in detail, and asking parents for any additional information.
- Police systematically coerce confessions from children.
- School's owners and staff were accused of ritually abusing of hundreds of children as part of their Satanic worship. All charges finally dropped after seven years.

Elizabeth Loftus and the Misinformation Effect

1993: "The reality of repressed memories" [22]

- "a rise in reported memories of childhood sexual abuse that were allegedly repressed for many years."
- "(a) How common is it for memories of child abuse to be repressed? (b) How are jurors and judges likely to react to these repressed memory claims? (c) When the memories surface, what are they like? and (d) How authentic are the memories?"

Theoretical critique

What it is

- The misinformation paradigm considers individuals making, reading, and sharing false statements, usually on social media, producing research that often warns of the harm to society resulting from the sum of many such events.

What it's not

- Poltical economy of social media like Parenti[2] [23] and Herman and Chomsky [24] did for mass media in 1986 and 1988.
- Media theory: Williams [25] brings together the technology of broadcasting and the cultural form of television to describe the "flow" state of watching television, simultaneously communal yet private

[2] Do people here know about Parenti?

Misinformation sees through the medium

"The COVID-19 Social Media Infodemic" [26] (all emphasis added)

"Social **media** platforms such as YouTube and Twitter provide **direct access** to an unprecedented amount of content and may amplify rumors and questionable information"

Compare to...

- Williams [25] described and theorized the design decisions that went into television
- Parenti [23] and Herman/Chomsky [24] described the political processes by which the news becomes unreliable

Misinformation studies rarely mention that social media is designed

Social media is not...

- trivial[3]
- inevitable
- naturally occurring
- technologically determined
- democratizing anything (e.g., "...ordinary people can engage in public speech and actions **without mediation by the mainstream media or other traditional sources of power.**" [27])

Social media is...

- designed, built, and actively maintained
- owned
- dynamic
- operated for profit
- politically contentious (e.g., Twitter/X) in both its use and design

[3] iykyk

Misinformation

- Despite emphasis on social media, the post-2016 misinformation paradigm looks less like media theory and more like the Loftus community's cognitive research.
- Focus on the content, sources, and consumers of misinformation, not its context, cultural form, or structural factors.
  - Resonance with social media database design, whose schema are also comprised of those same objects (future work!)

Misinformation

Origin (my claim)

- Change in emphasis from cognition to the misinformation itself, along with an increase in scale, but **the core model is recognizable** from Loftus community
- Caveat: This is one strand of many (mostly much smaller) strands that the paradigm weaves together

Theoretical Problem: No theory of medium

- Dead horse thoroughly beaten

Theoretical problem: Zombie laboratory model

- Misinformation concept was grounded in interactions between, for example, the police and victims

Theoretical problem: The truth does not live in the text

- "The pope is dead"
- "You see, Meno, that I am not teaching the boy anything, but all I do is question him" [28]

Practical/Theoretical problem: Lack of emergent theory

- Untenable commitment to evaluating individual pieces of misinformation
- Within paradigm, the whole is treated as the sum of its parts

Lack of emergent theory: The problems of scale

Researchers often define misinformation as a social media post that includes links to known unreliable sources [29] [30] [31] [32]

- "**Following previous research**, we used a domain-level approach to measurement" [32]
- "We annotate the scientific quality of retweets based on the web domains they reference—a measure that is also **commonly used in other research, rather than evaluating the content of the individual articles themselves.**" [31]

(all emphasis added)

Ignores anything that is not a direct link to an unreliable source

- Reliable/unreliable binary is untenable
- Ignores the spread within social media

Assumes that everything unreliable sources say is untrue

- Contradicts the documented best practices of those professionally employed to shape public opinion in both government and industry [33]

Example from circa 1940

"When there is no compelling reason to suppress a fact, tell it. [...] Aside from considerations of military security, the only reason to suppress a piece of news is if it is unbelievable [...] When the listener catches you in a lie, your power diminishes [...] For this reason, never tell a lie which can be discovered."

- Supreme Headquarters Allied Expeditionary Force of World War II [34]

An oversimplification exposes fundamental misunderstanding

- Public relations, military propaganda, intelligence agencies, etc. operate in the chasm between true and false, rather than at its poles
- The more likely that a piece of misinformation is to be believed, the more difficult it necessarily is to detect as misinformation
- **Untenable theoretical commitments force misinformation scholars to focus their inquiry precisely on the least salient part of the phenomenon**

Conclusion

- Misinformation papers constantly warn of democracy under threat, and of distrust in experts, science and/or institutions
- Some authors argue that misinformation and distrust in institutions are linked [35] , or that misinformation causes distrust in institutions [36]
- No one never asks whether science/experts/institutions have earned the public trust, or how they ought to behave in moments of contention for epistemic authority

Conclusion

Embrace humility

- Each person has been exposed to a lifetime of unending propaganda
- None of us are above it, nor can anyone easily spot all of it

Defend democracy

- If you believe democracy is under threat, do not treat the public like idiots needing to be managed, whose sole calling is to obey experts
- Participate in public discourse rather than research how to manage it for governments and companies
- Academics are teachers, not social media moderators

Fix institutions

- Our institutions are not worthy of public trust
- Academics are well positioned to exert power over their workplaces

Actually do science

- Science is the search for knowledge, not the counting of statements that disagree with official knowledge-dispensing institutions

Future work

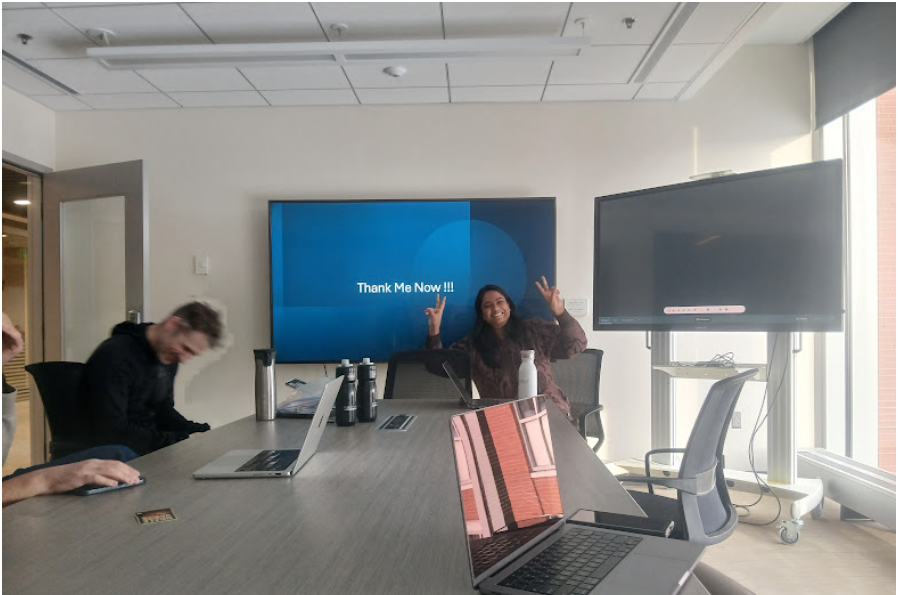
On misinformation (the concept)

- More detail on the origin of the paradigm
  - Talk to early paradigm adopters (examples in 2012 or earlier!)
  - Twitter
  - Mass media
  - Resonance with social media database design, whose schema parallels the structure of Loftus Research

More generally

- Why is this happening?

Thanks for listening! Thanks especially to all collaborators!



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